Introduction to Computer Vision (CS642)

Local Image Features - Lecture 05

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Contents

Feature Matching

Correspondence across views

- Correspondence is the matching of points, patches, edges, or regions across the images.

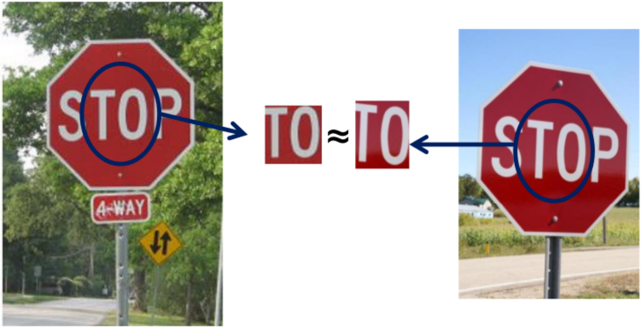


Figure: Reprinted from

Feature Matching

Example: Fundamental Matrix

- Estimating the fundamental matrix that corresponds two views.

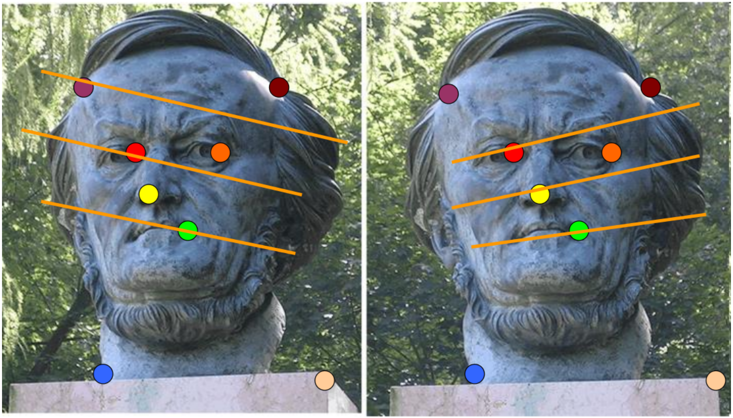


Figure: Reprinted from

- Estimating the fundamental matrix that corresponds two views.

Figure: Reprinted from

Contents

Interest points

Definition

- Interest point is the well defined position in the image.
- Interest point is also called key points or features.
- Applications:
  - Tracking: Which points are good to track?
  - Recognition: Find patches likely to tell us something about object category
  - 3D reconstruction: Find correspondences across different views.

- What points are easier to remember in the image and recalled after the image is deformed.

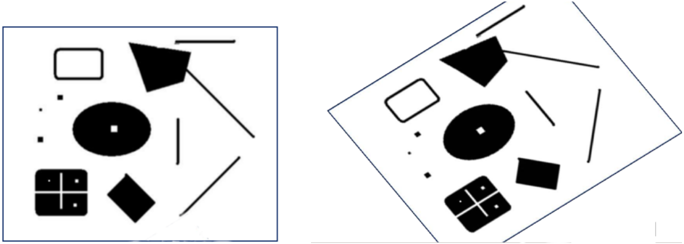
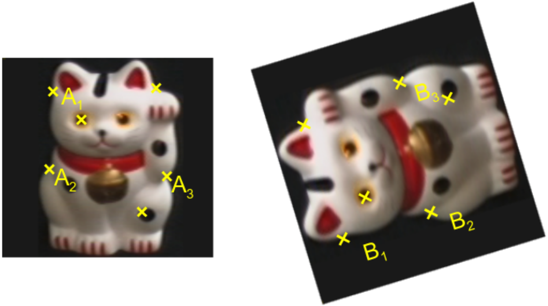


Table: Points to remember in the deformed image

Interest points
Key point matching

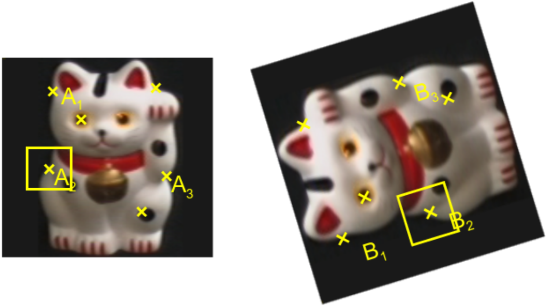
K. Grauman, B. Leibe

Interest points
Key point matching



1. **Find a set of distinctive key-points**

K. Grauman, B. Leibe

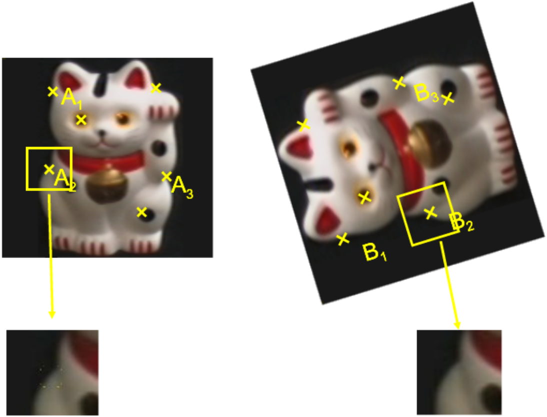


1. **Find a set of distinctive key-points**
2. **Define a region around each keypoint**

K. Grauman, B. Leibe

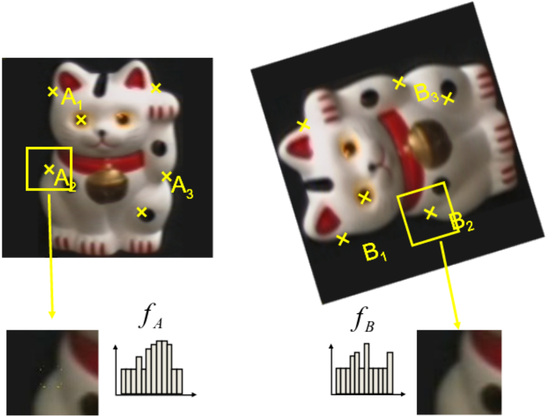
Interest points

Key point matching



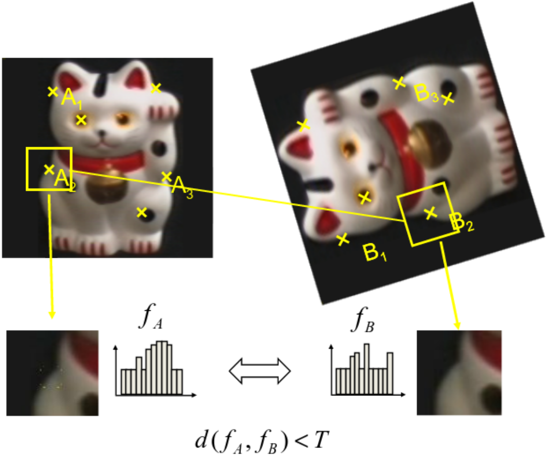
1. Find a set of distinctive key-points
2. Define a region around each keypoint
3. Extract and normalize the region content

K. Grauman, B. Leibe



1. **Find a set of distinctive key-points**
2. **Define a region around each keypoint**
3. **Extract and normalize the region content**
4. **Compute a local descriptor from the normalized region**

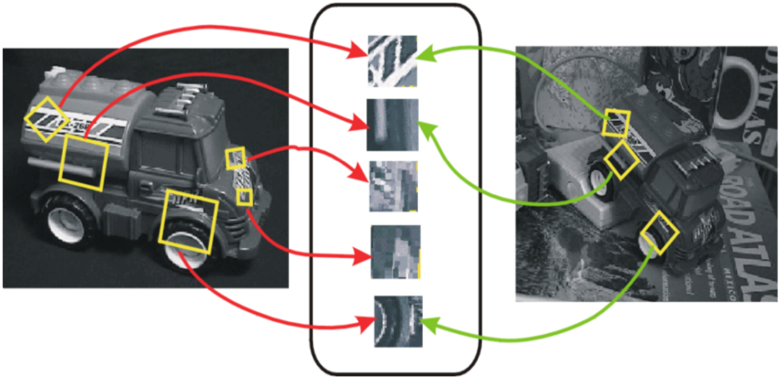
K. Grauman, B. Leibe



1. **Find a set of distinctive key-points**
2. **Define a region around each keypoint**
3. **Extract and normalize the region content**
4. **Compute a local descriptor from the normalized region**
5. **Match local descriptors**

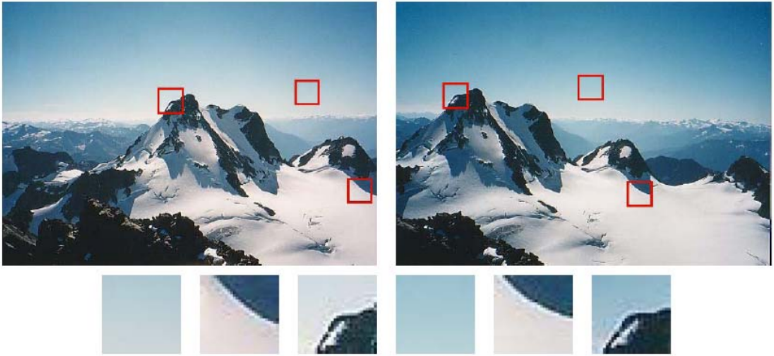
K. Grauman, B. Leibe

- Image content is transformed into local feature coordinates that are invariant to translation, rotation, scale, and other imaging parameters



Interest points

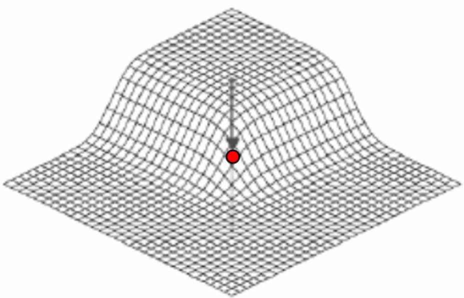
- Some patches can be localized or matched with higher accuracy than others.



Contents

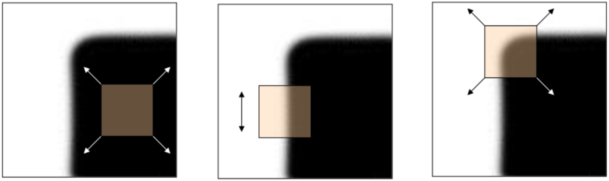
Interest point detection

- Expressive texture
  - The point at which the direction of the boundary of object changes abruptly
  - Intersection point between two or more edge segments



Interest point detection

- We should easily recognize the point by looking through a small window
- Shifting a window in any direction should give a large change in intensity



(a) (b) (c)

Table: (a) flat region: no change in all directions. (b) edge: no change along the edge direction. (c) corner: significant change in all directions

- Detect all (or most) true interest points
- No false interest points
- Well localized.
- Robust with respect to noise.
- Efficient detection

- Based on brightness of images
  - Usually image derivatives
- Based on boundary extraction
  - First step edge detection
  - Curvature analysis of edges

- Corner point can be recognized in a window
- Shifting a window in any direction should give a large change in intensity

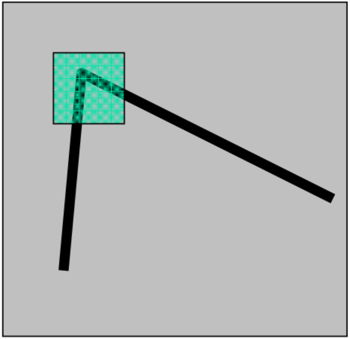
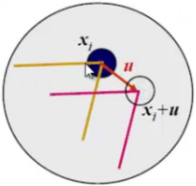
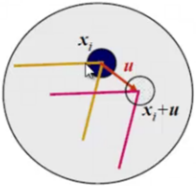


Figure: Reprinted from Harris, M. Stephens, 1988.



(a)

Interest point detection
Aperture Problem



(a)

(b)

Interest point detection
Aperture Problem

(a) (b) (c)

Contents

Harris Interest Point Detector

Correlation

$$f \otimes h = \sum_{k} \sum_{j} f(k,l)h(i+k, j+l)$$

$$f = \text{Image}$$
$$h = \text{Kernel}$$

$$\overset{f}{\begin{bmatrix} f_1 & f_2 & f_3 \\ f_4 & f_5 & f_6 \\ f_7 & f_8 & f_9 \end{bmatrix}} \otimes \overset{h}{\begin{bmatrix} h_1 & h_2 & h_3 \\ h_4 & h_5 & h_6 \\ h_7 & h_8 & h_9 \end{bmatrix}} \longrightarrow \begin{aligned} f * h = {} & f_1h_1 + f_2h_2 + f_3h_3 \\ & + f_4h_4 + f_5h_5 + f_6h_6 \\ & + f_7h_7 + f_8h_8 + f_9h_9 \end{aligned}$$

- Cross Correlation

$$f \otimes h = \sum_{k} \sum_{l} f(k,l)h(i+k, j+l)$$

- Auto Correlation

$$f \otimes h = \sum_{k} \sum_{l} f(k,l)f(i+k, j+l)$$

- Sum of Square Differences (SSD)

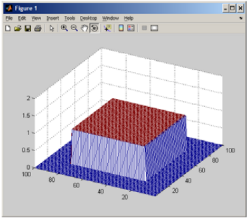
$$SSD = \sum_{k} \sum_{l} (f(k,l) - h(i+k, j+l))^2$$

Harris Interest Point Detector

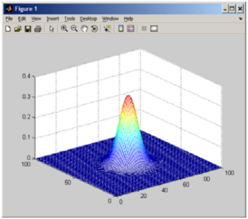
- Change of intensity for the shift (u,v) based on auto correlation.

$$E(u,v) = \sum_{x,y} w(x,y)\left[\mathtt{I}(x+u,y+v) - \mathtt{I}(x,y)\right]^2$$

- Window function



UNIFORM



GAUSSIAN

$$E(u,v)=\sum_{x,y}\underbrace{w(x,y)}_{\text{window function}}\,[\underbrace{I(x+u,y+v)}_{\text{shifted intensity}}-\underbrace{I(x,y)}_{\text{intensity}}]^2$$

$$E(u,v)=\sum_{x,y}\underbrace{w(x,y)}_{\text{window function}}\,[\underbrace{I(x,y)+uI_x+vI_y}_{\text{shifted intensity}}-\underbrace{I(x,y)}_{\text{intensity}}]^2$$

Taylor Series

$$E(u,v)=\sum_{x,y}w(x,y)[uI_x+vI_y]^2$$

$$E(u,v)=\sum_{x,y}w(x,y)\left[\begin{pmatrix}u & v\end{pmatrix}\begin{pmatrix}I_x\\ I_y\end{pmatrix}\right]^2$$

$$E(u,v)=\sum_{x,y}w(x,y)\begin{pmatrix}u & v\end{pmatrix}\begin{pmatrix}I_x\\ I_y\end{pmatrix}\begin{pmatrix}I_x & I_y\end{pmatrix}\begin{pmatrix}u\\ v\end{pmatrix}$$

$$E(u,v)=\begin{pmatrix}u & v\end{pmatrix}\left[\sum_{x,y}w(x,y)\begin{pmatrix}I_x\\ I_y\end{pmatrix}\begin{pmatrix}I_x & I_y\end{pmatrix}\right]\begin{pmatrix}u\\ v\end{pmatrix}$$

$$E(u,v)=\begin{pmatrix}u & v\end{pmatrix}M\begin{pmatrix}u\\ v\end{pmatrix}$$

Slide Source: Mubarak Shah

Mathematics of Harris Interest Point Detector

$$E(u,v) = \begin{bmatrix} u & v \end{bmatrix} \mathtt{M} \begin{bmatrix} u \\ v \end{bmatrix}$$

where,

$$\mathtt{M} = \sum_{x,y} w(x,y) \begin{bmatrix} \mathtt{I}_x\mathtt{I}_x & \mathtt{I}_x\mathtt{I}_y \\ \mathtt{I}_x\mathtt{I}_y & \mathtt{I}_y\mathtt{I}_y \end{bmatrix}$$

We can write the equation without weight

$$\mathtt{M} = \begin{bmatrix} \sum \mathtt{I}_x\mathtt{I}_x & \sum \mathtt{I}_x\mathtt{I}_y \\ \sum \mathtt{I}_x\mathtt{I}_y & \sum \mathtt{I}_y\mathtt{I}_y \end{bmatrix} = \sum \left(\begin{bmatrix} \mathtt{I}_x \\ \mathtt{I}_y \end{bmatrix} \begin{bmatrix} \mathtt{I}_x & \mathtt{I}_y \end{bmatrix} \right) = \sum \nabla \mathtt{I} (\nabla \mathtt{I})^T$$

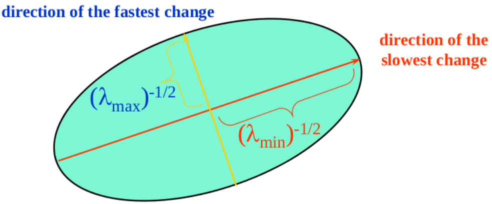
Mathematics of Harris Interest Point Detector

$$E(u,v) = \begin{bmatrix} u & v \end{bmatrix} \mathtt{M} \begin{bmatrix} u \\ v \end{bmatrix}$$

where,

$$\mathtt{M} = \sum_{x,y} w(x,y) \begin{bmatrix} \mathtt{I}_x^2 & \mathtt{I}_x\mathtt{I}_y \\ \mathtt{I}_x\mathtt{I}_y & \mathtt{I}_y^2 \end{bmatrix}$$

- $E(u,v)$ is an equation of an ellipse, where $\mathtt{M}$ is the covariance.
- Let λ_1 and λ_2 be eigenvalues of $\mathtt{M}$



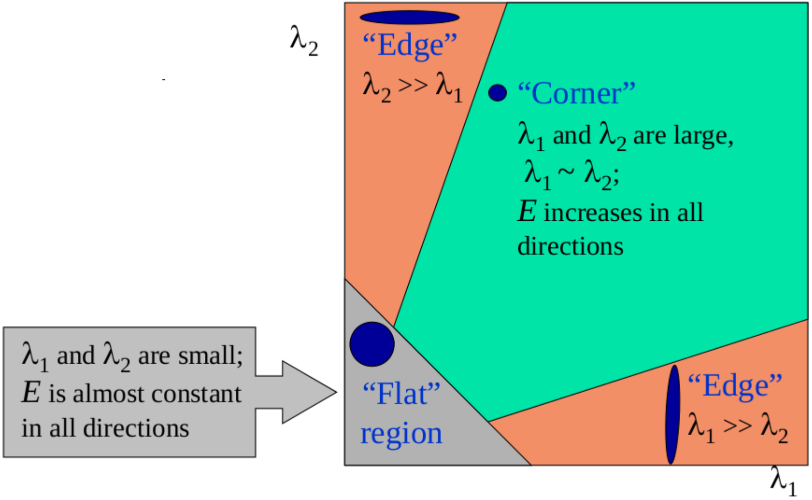
Mathematics of Harris Interest Point Detector

$$E(u, v) = \begin{bmatrix} u & v \end{bmatrix} \mathtt{M} \begin{bmatrix} u \\ v \end{bmatrix}$$

where,

$$\mathtt{M} = \sum_{x,y} w(x, y) \begin{bmatrix} \mathtt{I}_x^2 & \mathtt{I}_x \mathtt{I}_y \\ \mathtt{I}_x \mathtt{I}_y & \mathtt{I}_y^2 \end{bmatrix} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$$

- Classification of image points using eigenvalues of M:



Harris Interest Point Detector

- Measure of cornerness in terms of λ_1 , λ_2

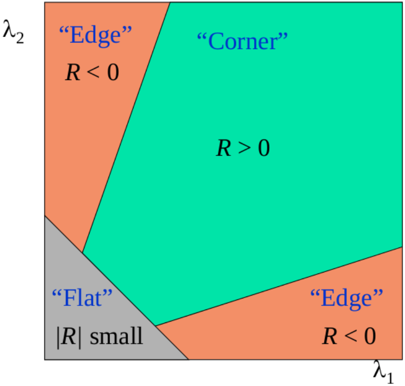
$$\mathtt{M} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$$

$$R = \det(\mathtt{M}) - k[trace(\mathtt{M})]^2$$
$$R = \lambda_1\lambda_2 - k(\lambda_1 + \lambda_2)^2$$

where k is a constant ranges 0.04 to 0.06.

Harris Interest Point Detector

- R is large for the interest point.
- R is negative with large magnitude for an edge.
- $|R|$ is small for a flat region.



Harris Interest Point Detector

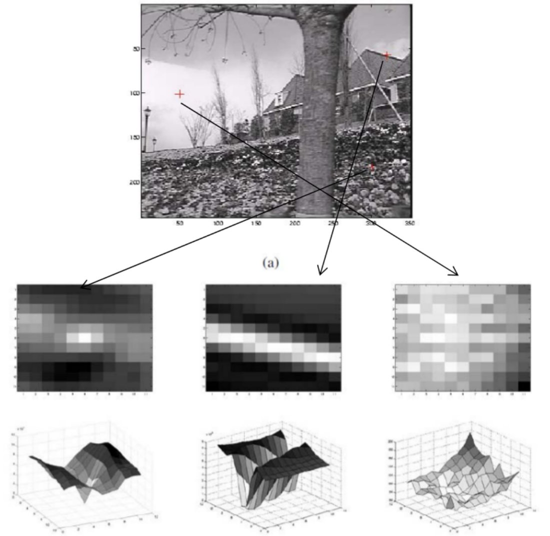


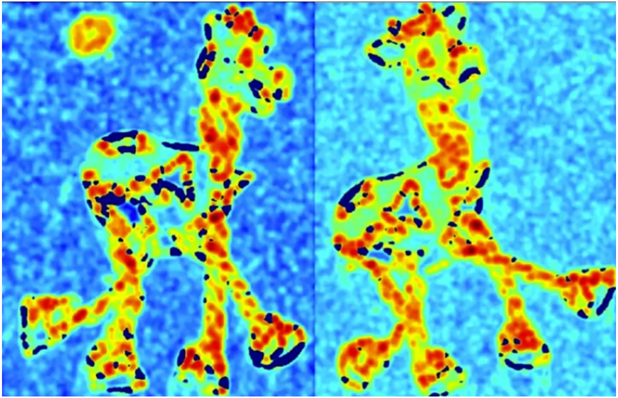
Figure: (a) Interest point, (b) Edges, and (c) flat region.

- Compute horizontal and vertical derivatives $\mathtt{I}_x$ and $\mathtt{I}_y$ of the image.
- Compute three images corresponding to three terms $\mathtt{I}_x^2$, $\mathtt{I}_y^2$, and $\mathtt{I}_x\mathtt{I}_y$ in matrix $\mathtt{M}$.
- Convolve the three images with a larger Gaussian filter.
- Compute scalar cornerness value using one of the R measures.
- Find local maxima above some threshold as detected interest points.

Harris Interest Point Detector

- Consider a toy giraffe to apply Harris algorithm.

- Compute corner response R.

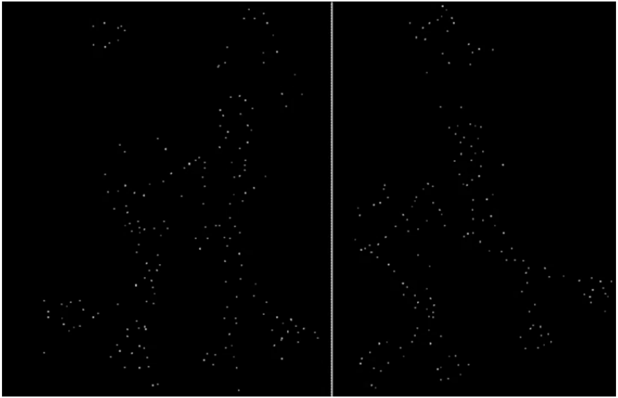


Harris Interest Point Detector

- Find points with larger corner response $R >$ threshold.

Harris Interest Point Detector

- Take only the points of local maxima of R.



Contents

Eigen Vectors and Eigen Values

- The Eigen vector $\boldsymbol{x}$ of a matrix $\mathtt{A}$ is a special vector with the following property

$$\mathtt{A}\boldsymbol{x} = \lambda\boldsymbol{x}$$

The End